

BACH LONG VI, Viet Nam

WMO: 488390

Lat: 20.130N

Lon: 107.720E

Elev: 56

StdP: 100.65

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	11.3	12.2	4.5	5.2	14.3	6.3	6.0	14.4	17.2	13.7	16.1	13.6	11.0	20	0.512

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	3.5	33.0	28.7	32.4	28.5	31.8	28.2	29.2	32.1	28.8	31.5	28.5	31.1	5.6	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
28.4	24.8	31.3	28.1	24.4	30.9	27.7	23.9	30.5	96.3	32.4	94.4	31.9	92.9	31.3	31.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
16.2	14.2	12.5	DB	10.7	34.6	1.3	1.6	9.7	35.8	8.9	36.7	8.2	37.6	7.2	38.7	(4)
			WB	8.0	29.9	1.3	0.5	7.0	30.3	6.2	30.6	5.5	30.9	4.5	31.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	24.4	17.7	18.0	19.8	23.9	27.3	29.4	29.5	29.0	28.0	26.2	23.3	19.8
	DBStd	4.94	3.12	3.56	3.00	2.81	1.88	1.32	1.40	1.34	1.42	1.69	2.55	2.81
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	137	51	47	19	1	0	0	0	0	0	0	2	17
	CDD10.0	5238	238	223	304	416	537	582	604	589	539	503	401	303
	CDD18.3	2334	31	37	66	167	278	332	346	331	289	244	152	62
	CDH23.3	22608	16	49	131	946	2778	4182	4406	4067	3217	2043	665	108
	CDH26.7	7762	0	1	5	142	806	1820	1961	1635	984	331	72	7
Wind	WSAvg	6.1	6.7	6.2	5.7	5.4	5.7	6.5	6.6	5.0	5.2	5.9	6.7	7.3
Precipitation	PrecAvg	1168	28	18	30	39	104	158	166	259	226	87	42	24
	PrecMax	1711	282	88	93	102	339	344	543	727	466	218	173	71
	PrecMin	729	1	0	2	0	13	7	26	98	39	0	0	1
	PrecStd	256	55	23	25	31	79	95	121	145	109	63	40	22
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	25.0	26.3	27.3	30.7	32.5	33.8	33.8	33.3	32.6	31.0	29.8	27.5
		MCWB	22.1	23.4	24.5	26.9	28.5	28.8	29.0	28.8	28.9	26.7	25.8	24.5
	2%	DB	23.6	24.9	26.0	29.2	31.6	32.8	33.1	32.6	31.6	30.0	28.5	25.8
		MCWB	21.1	22.7	23.6	26.1	28.2	28.7	28.7	28.5	28.2	26.5	25.3	23.3
	5%	DB	22.4	23.6	24.8	28.2	30.8	32.1	32.4	31.8	30.8	29.2	27.3	24.5
		MCWB	20.6	21.8	23.0	25.6	27.8	28.4	28.3	28.2	27.6	25.9	24.4	22.5
	10%	DB	21.4	22.5	23.7	27.3	30.0	31.5	31.7	31.1	30.2	28.4	26.2	23.3
		MCWB	20.0	21.2	22.3	25.1	27.4	28.2	28.0	27.9	27.1	25.4	23.8	21.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	22.6	23.8	24.8	27.2	29.2	29.6	29.6	29.4	29.2	27.7	26.6	25.1
		MCDB	24.4	25.9	26.7	29.8	31.6	32.3	33.0	32.5	31.9	30.0	29.0	26.9
	2%	WB	21.6	22.9	23.9	26.5	28.5	29.1	29.0	28.9	28.5	26.9	25.7	23.9
		MCDB	23.1	24.5	25.5	28.7	30.9	31.9	32.2	31.8	30.9	29.2	27.8	25.1
	5%	WB	20.9	22.1	23.2	25.9	28.0	28.7	28.6	28.5	28.0	26.4	25.1	23.0
		MCDB	22.1	23.3	24.5	27.8	30.2	31.3	31.6	31.1	30.2	28.5	26.7	24.4
	10%	WB	20.2	21.4	22.5	25.4	27.6	28.4	28.2	28.1	27.5	25.9	24.5	21.9
		MCDB	21.2	22.4	23.7	27.0	29.6	30.8	31.0	30.6	29.7	27.9	26.0	23.0
Mean Daily Temperature Range	5% DB	MDBR	3.2	3.3	3.5	4.0	4.0	3.6	3.5	3.6	3.4	3.1	3.2	3.1
		MCDBR	4.8	5.0	5.1	5.0	4.6	4.4	4.3	4.3	4.2	4.1	4.6	5.0
		MCWBR	3.1	3.0	2.9	2.6	2.4	2.1	2.1	2.3	2.6	2.5	2.7	3.1
	5% WB	MCDBR	4.7	4.7	4.8	4.9	4.4	4.0	4.0	4.0	4.0	3.8	4.4	4.9
MCWBR		3.3	3.0	3.0	2.6	2.5	2.2	2.2	2.4	2.6	2.5	2.8	3.3	
Clear-Sky Solar Irradiance	taub	0.622	0.651	0.790	0.731	0.536	0.495	0.473	0.519	0.574	0.675	0.599	0.636	
	taud	1.735	1.696	1.495	1.636	2.144	2.281	2.340	2.185	1.982	1.706	1.840	1.718	
	Ebn at Noon	658	671	601	648	781	806	825	793	744	644	672	627	
	Edh at Noon	216	236	299	262	156	135	128	150	181	231	193	214	
All-Sky Solar Radiation	RadAvg	2.52	2.98	3.31	4.47	5.41	5.44	5.35	4.93	4.69	4.36	3.54	2.72	
	RadStd	0.57	0.51	0.43	0.48	0.45	0.51	0.63	0.33	0.42	0.29	0.42	0.41	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	+0.50	+0.38	+0.38	N/A	N/A	N/A	N/A
(47)	Regional (4 neighbors)	+0.23	N/A	N/A	+0.37	+0.25	+0.28	N/A	N/A	+75	+75

Nomenclature: See separate page